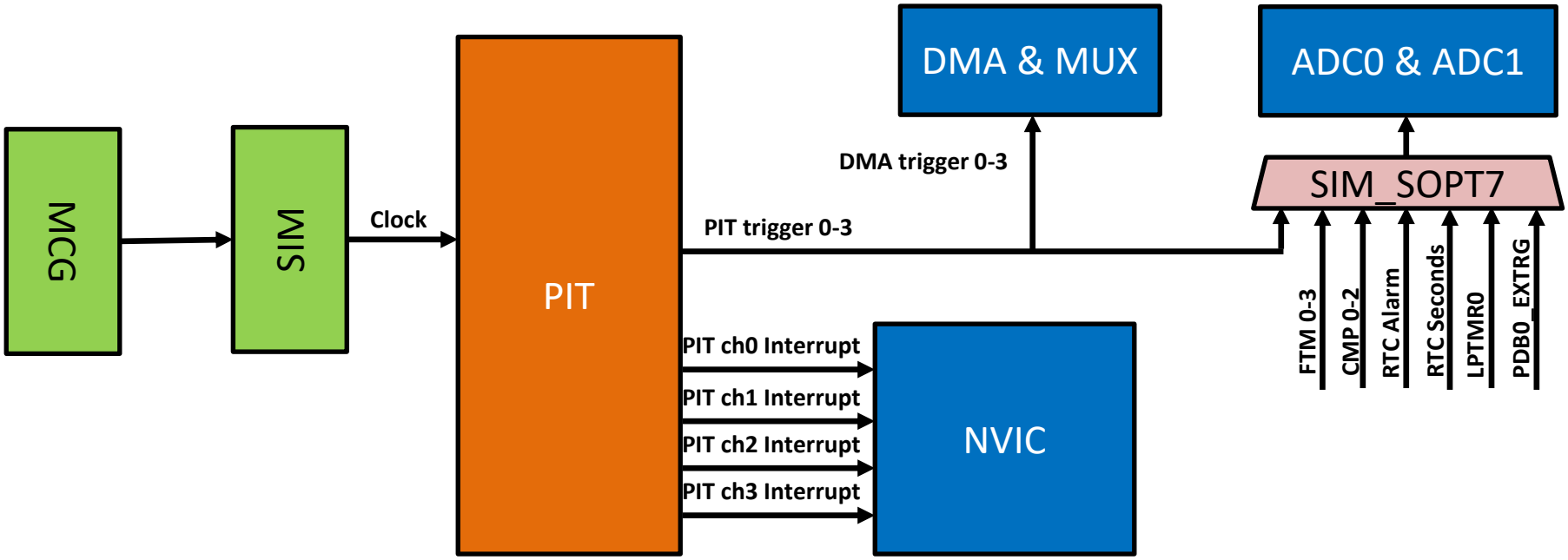
PIT
Periodic Interrupt Timer



PIT: **P**rogramable **I**nterval **T**imer



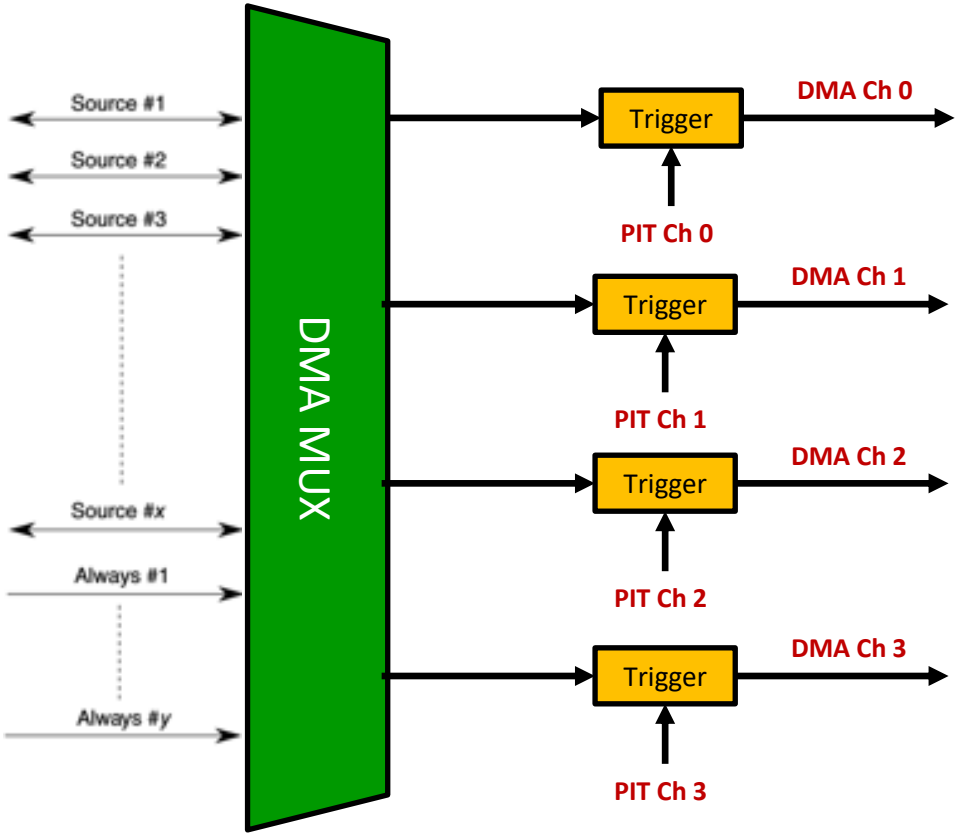
El Modulo PIT consiste en un conjunto de timers usados para:

- Generar interrupciones periódicas
- Proveer Triggers periódicos para el modulo de DMA y ADC

PIT to DMA Trigger

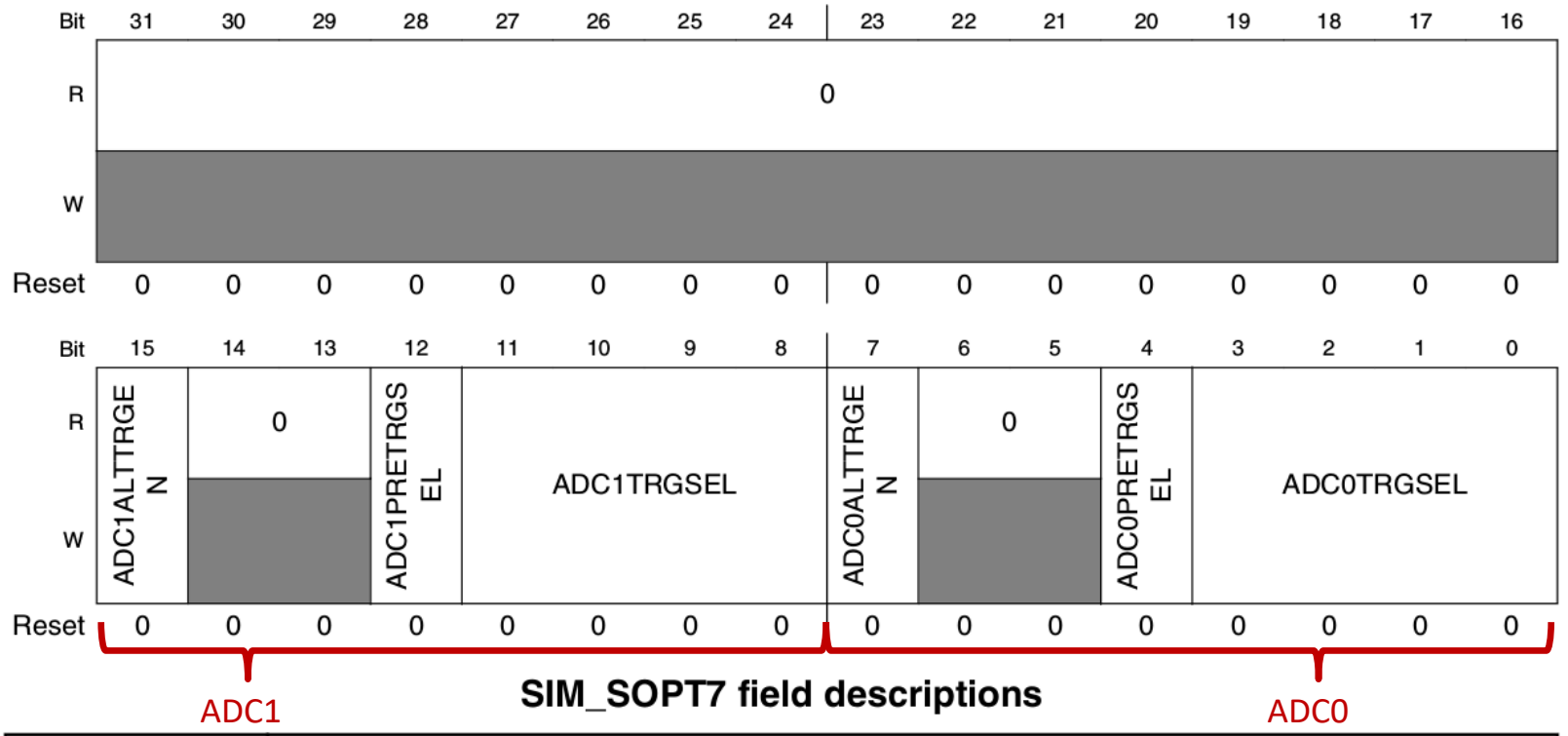
Table 3-61. PIT channel assignments for periodic DMA triggering

DMA Channel Number	PIT Channel
DMA Channel 0	PIT Channel 0
DMA Channel 1	PIT Channel 1
DMA Channel 2	PIT Channel 2
DMA Channel 3	PIT Channel 3



PIT to ADC Trigger (SIM_SOPT7)

Address: 4004_7000h base + 1018h offset = 4004_8018h



SIM_SOPT7 field descriptions

Field	Description
31–16 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.

PIT to ADC0 Trigger

Field	Description
7 ADC0ALTTRGEN	ADC0 alternate trigger enable Enable alternative conversion triggers for ADC0. 0 PDB trigger selected for ADC0. 1 Alternate trigger selected for ADC0.
6–5 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
4 ADC0PRETRGSEL	ADC0 pretrigger select Selects the ADC0 pre-trigger source when alternative triggers are enabled through ADC0ALTTRGEN. 0 Pre-trigger A 1 Pre-trigger B
ADC0TRGSEL	ADC0 trigger select
	Selects the ADC0 trigger source when alternative triggers are functional in stop and VLPS modes. . 0000 PDB external trigger pin input (PDB0_EXTRG) 0001 High speed comparator 0 output 0010 High speed comparator 1 output 0011 High speed comparator 2 output 0100 PIT trigger 0 0101 PIT trigger 1 0110 PIT trigger 2 0111 PIT trigger 3 1000 FTM0 trigger 1001 FTM1 trigger 1010 FTM2 trigger 1011 FTM3 trigger 1100 RTC alarm 1101 RTC seconds 1110 Low-power timer (LPTMR0) trigger 1111 Reserved

PIT to ADC1 Trigger

SIM_SOPT7 field descriptions (continued)

Field	Description
15 ADC1ALTTRGEN	ADC1 alternate trigger enable Enable alternative conversion triggers for ADC1. 0 PDB trigger selected for ADC1 1 Alternate trigger selected for ADC1 as defined by ADC1TRGSEL.
14–13 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
12 ADC1PRETRGSEL	ADC1 pre-trigger select Selects the ADC1 pre-trigger source when alternative triggers are enabled through ADC1ALTTRGEN. 0 Pre-trigger A selected for ADC1. 1 Pre-trigger B selected for ADC1.
11–8 ADC1TRGSEL	ADC1 trigger select Selects the ADC1 trigger source when alternative triggers are functional in stop and VLPS modes. 0000 PDB external trigger pin input (PDB0_EXTRG) 0001 High speed comparator 0 output 0010 High speed comparator 1 output 0011 High speed comparator 2 output 0100 PIT trigger 0 0101 PIT trigger 1 0110 PIT trigger 2 0111 PIT trigger 3 1000 FTM0 trigger 1001 FTM1 trigger 1010 FTM2 trigger 1011 FTM3 trigger 1100 RTC alarm 1101 RTC seconds 1110 Low-power timer (LPTMR) trigger 1111 Reserved

PIT Clocking

PIT Clock = Bus Clock[1]

Address: 4004_7000h base + 103Ch offset = 4004_803Ch

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	DAC0	1	RTC	0	ADC0	FTM2	FTM1	FTM0	PIT	PDB	USBDCD	0		CRC	0	
W																
Reset	0	1	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	I2S0	0	SPI1	SPI0	0		RNGA	0				FLEXCAN0	0		DMAMUX	FTF
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	1

SIM_SCGC6 field descriptions

```
SIM->SCGC6 |= SIM_SCGC6_PIT_MASK;
```

1-bus clock = core clock/2 =100 MHz/2 =50MHz en nuestro caso

PIT Module Control Register: MCR

PIT_MCR field descriptions

Field	Description
31–3 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
2 Reserved	This field is reserved.
1 MDIS	Module Disable - (PIT section) Disables the standard timers. This field must be enabled before any other setup is done. 0 Clock for standard PIT timers is enabled. 1 Clock for standard PIT timers is disabled.
0 FRZ	Freeze Allows the timers to be stopped when the device enters the Debug mode. 0 Timers continue to run in Debug mode. 1 Timers are stopped in Debug mode.

PIT
- PIT_MCR
- PIT_LDVALn
- PIT_CVALn
- PIT_TCTRLn
- PIT_TFLGn

PIT Timer Load Value Register: LDVALn

Address: 4003_7000h base + 100h offset + (16d × i), where i=0d to 3d

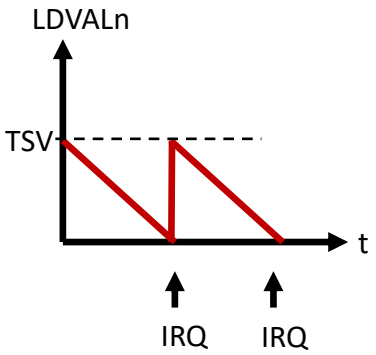
Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R W	TSV																															
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PIT_LDVALn field descriptions

Field	Description
TSV	Timer Start Value Sets the timer start value. The timer will count down until it reaches 0, then it will generate an interrupt and load this register value again. Writing a new value to this register will not restart the timer; instead the value will be loaded after the timer expires. To abort the current cycle and start a timer period with the new value, the timer must be disabled and enabled again.

PIT
- PIT_MCR
- PIT_LDVALn
- PIT_CVALn
- PIT_TCTRLn
- PIT_TFLGn

This register select the timeout period for the timer interrupts

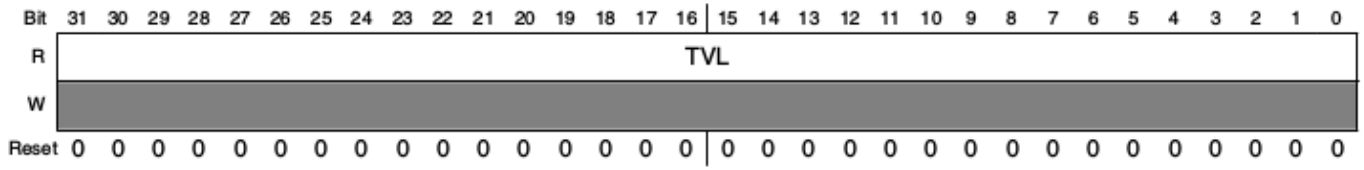


PIT Current Timer Value Register: CVALn

These registers indicate the current timer position.

Access: User read only

Address: 4003_7000h base + 104h offset + (16d × i), where i=0d to 3d



PIT_CVAL*n* field descriptions

Field	Description
TVL	Current Timer Value Represents the current timer value, if the timer is enabled. **NOTE:** • If the timer is disabled, do not use this field as its value is unreliable. • The timer uses a downcounter. The timer values are frozen in Debug mode if MCR[FRZ] is set.

PIT

- PIT_MCR
- PIT_LDVALn
- PIT_CVALn
- PIT_TCTRLn
- PIT_TFLGn

PIT Timer Control Register: TCTRLn

These registers contain the control bits for each timer.

Access: User read/write

Address: 4003_7000h base + 108h offset + (16d × i), where i=0d to 3d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

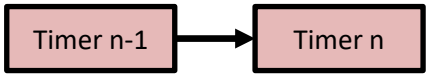
Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0													CHN	TIE	TEN
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PIT_TCTRL*n* field descriptions

Field	Description
31–3 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
2 CHN	Chain Mode When activated, Timer n-1 needs to expire before timer n can decrement by 1. Timer 0 cannot be chained. 0 Timer is not chained. 1 Timer is chained to previous timer. For example, for Channel 2, if this field is set, Timer 2 is chained to Timer 1.
1 TIE	Timer Interrupt Enable When an interrupt is pending, or, TFLGn[TIF] is set, enabling the interrupt will immediately cause an interrupt event. To avoid this, the associated TFLGn[TIF] must be cleared first. 0 Interrupt requests from Timer n are disabled. 1 Interrupt will be requested whenever TIF is set.
0 TEN	Timer Enable Enables or disables the timer. 0 Timer n is disabled. 1 Timer n is enabled.

PIT
- PIT_MCR
- PIT_LDVALn
- PIT_CVALn
- **PIT_TCTRLn**
- PIT_TFLGn

Chain Mode=1



PIT Timer Flag Register: TFLGn

Address: 4003_7000h base + 10Ch offset + (16d × i), where i=0d to 3d

Bit	31	30	29	28	27	26	25	24	23	22	21	20	19	18	17	16
R	0															
W																
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

Bit	15	14	13	12	11	10	9	8	7	6	5	4	3	2	1	0
R	0															TIF
W																w1c
Reset	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0	0

PIT
- PIT_MCR
- PIT_LDVALn
- PIT_CVALn
- PIT_TCTRLn
- PIT_TFLGn

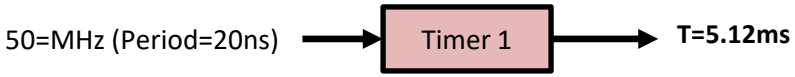
PIT_TFLG*n* field descriptions

Field	Description
31–1 Reserved	This field is reserved. This read-only field is reserved and always has the value 0.
0 TIF	Timer Interrupt Flag Sets to 1 at the end of the timer period. Writing 1 to this flag clears it. Writing 0 has no effect. If enabled, or, when TCTRLn[TIE] = 1, TIF causes an interrupt request. 0 Timeout has not yet occurred. 1 Timeout has occurred.

Ejemplos

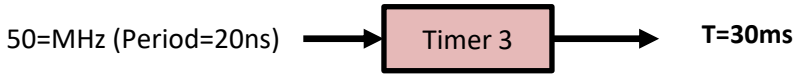
- The PIT clock has a frequency of 50 MHz.
- Timer 1 creates an interrupt every 5.12 ms.
- Timer 3 creates a trigger event every 30 ms.

PIT Clock = Bus Clock=50MHz



5.12 ms/20 ns = 256,000 cycles

```
PIT_LDVAL1=256000-1;  // setup Timer 1 for 256000 cycles
PIT_TCTRL1 = TIE;     // enable Timer 1 interrupts
PIT_TCTRL1 |= TEN;    // start Timer 1
```

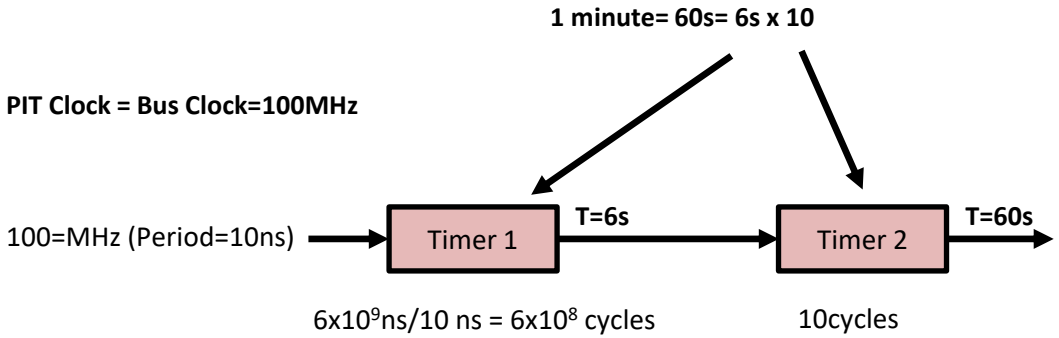


30 ms/20 ns = 1,500,000 cycles

```
PIT_LDVAL3=1.500.000-1;  // setup Timer 3 for 1.500.00 cycles
PIT_TCTRL3 = TIE;        // enable Timer3 interrupts
PIT_TCTRL3 |= TEN;       // start Timer3
```

Ejemplo chained timers

- The PIT clock has a frequency of 100 MHz.
- Timers 1 and 2 are available.
- An interrupt shall be raised every 1 minute.

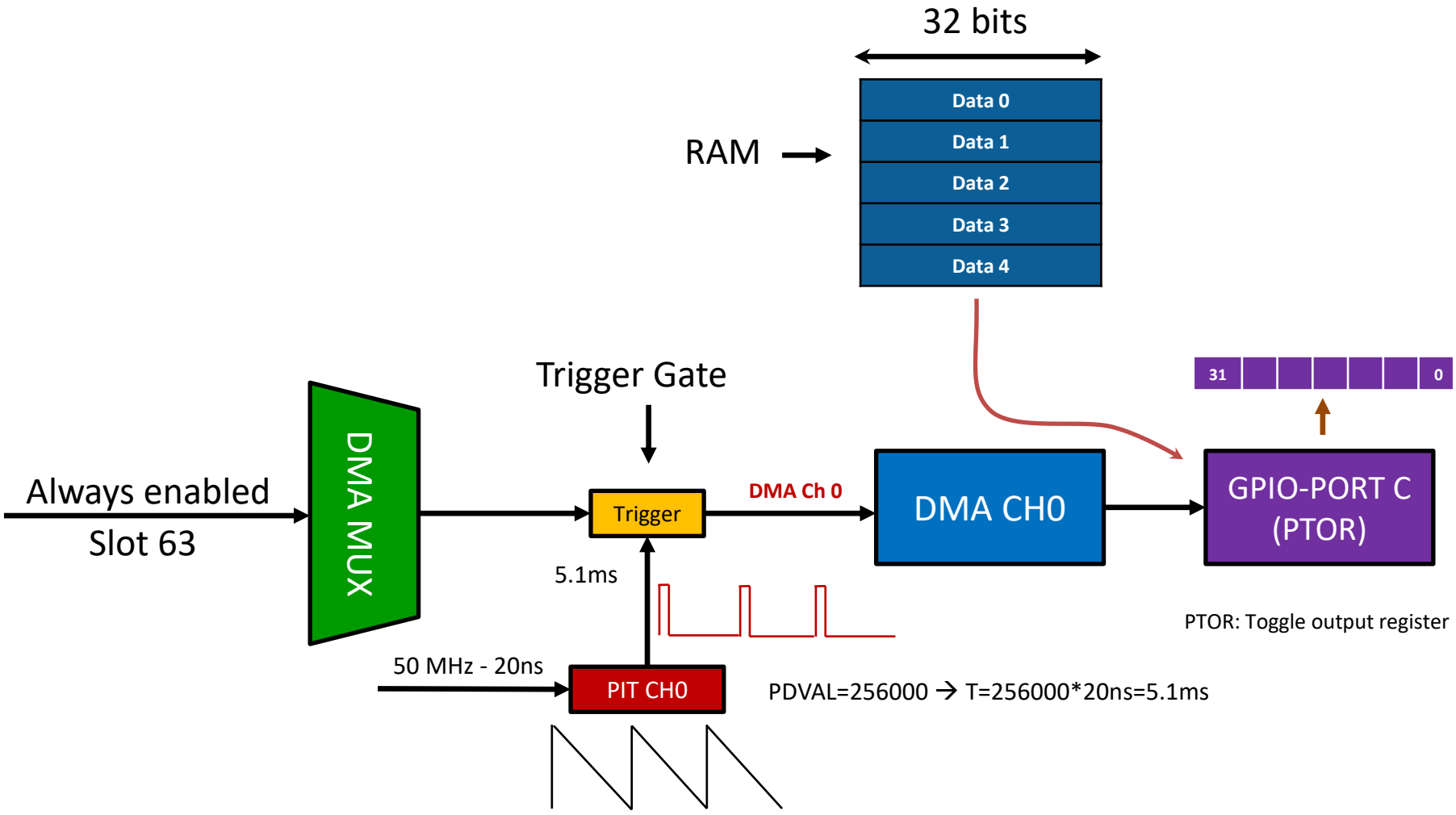



```
// Turn on PIT
PIT_MCR=0x00;

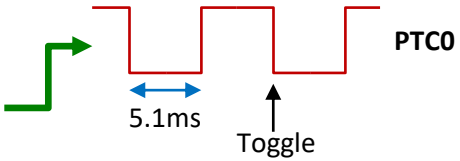
// Timer 2
PIT_LDVAL2=10-1  //Set up timer 2 for 10 counts
PIT_TCTRL2=TIE   //Enable Timer 2 interrupt
PIT_TCTRL2|=CHN  //Chain Timer 2 to Timer 1
PIT_TCTRL2|=TEN  //Start Timer 2

// Timer 1
PIT_LDVAL1=6x10^8   //Set up timer 1 for 600.000.000 counts
PIT_TCTRL1=TEN    //Start Timer 1
```

DMA – PIT Example



DMA Configuration




```
//List of Values to be written to the selected port
static uint32_t sourceBuffer[] = {0x0001,0x0001,0x0001,0x0001};
// PIT as Trigger , Always Enabled as DMA Request
DMAMUX->CHCFG[0] |= DMAMUX_CHCFG_ENBL_MASK | DMAMUX_CHCFG_TRIG_MASK | DMAMUX_CHCFG_SOURCE(63);
```

```
/* Set memory address for source and destination. */
DMA0->TCD[0].SADDR = (uint32_t)(sourceBuffer);      //List of Values to be written to the selected port
DMA0->TCD[0].DADDR = (uint32_t)(&(PTC->PTOR));  // Toggle PIN (only if '1' is written)
/* Set an offset for source and destination address. */
DMA0->TCD[0].SOFF =0x04; // Source address offset of 4 bytes per transaction.
DMA0->TCD[0].DOFF =0x00; // Destination address offset is 0. (Siempre al mismo lugar -> PTOR)
/* Set source and destination data transfer size to 32 bits (GPIO PTOR is 32 bits wide). */
DMA0->TCD[0].ATTR = DMA_ATTR_SSIZE(2) | DMA_ATTR_DSIZE(2);
/*Number of bytes to be transferred in each service request of the channel.*/
DMA0->TCD[0].NBYTES_MLNO =4; //1x4
/* Autosize for Current major iteration count */
DMA0->TCD[0].CITER_ELINKNO = DMA_CITER_ELINKNO_CITER((sizeof(sourceBuffer)/sizeof(sourceBuffer[0])));
DMA0->TCD[0].BITER_ELINKNO = DMA_BITER_ELINKNO_BITER((sizeof(sourceBuffer)/sizeof(sourceBuffer[0])));
/* Autosize SLAST for Wrap Around. This value is added to SADD at the end of Major Loop */
DMA0->TCD[0].SLAST = -sizeof(sourceBuffer);
/* Enable request signal for channel 0. */
DMA0->ERQ = DMA_ERQ_ERQ0_MASK;
```